INCH-POUND
MIL-C-55302/110F
31 October 1988
SUPERSEDING
MIL-C-55302/110E
10 August 1984

MILITARY SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, RIGHT ANGLE, 30 THRU 140 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SQ. GRID)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

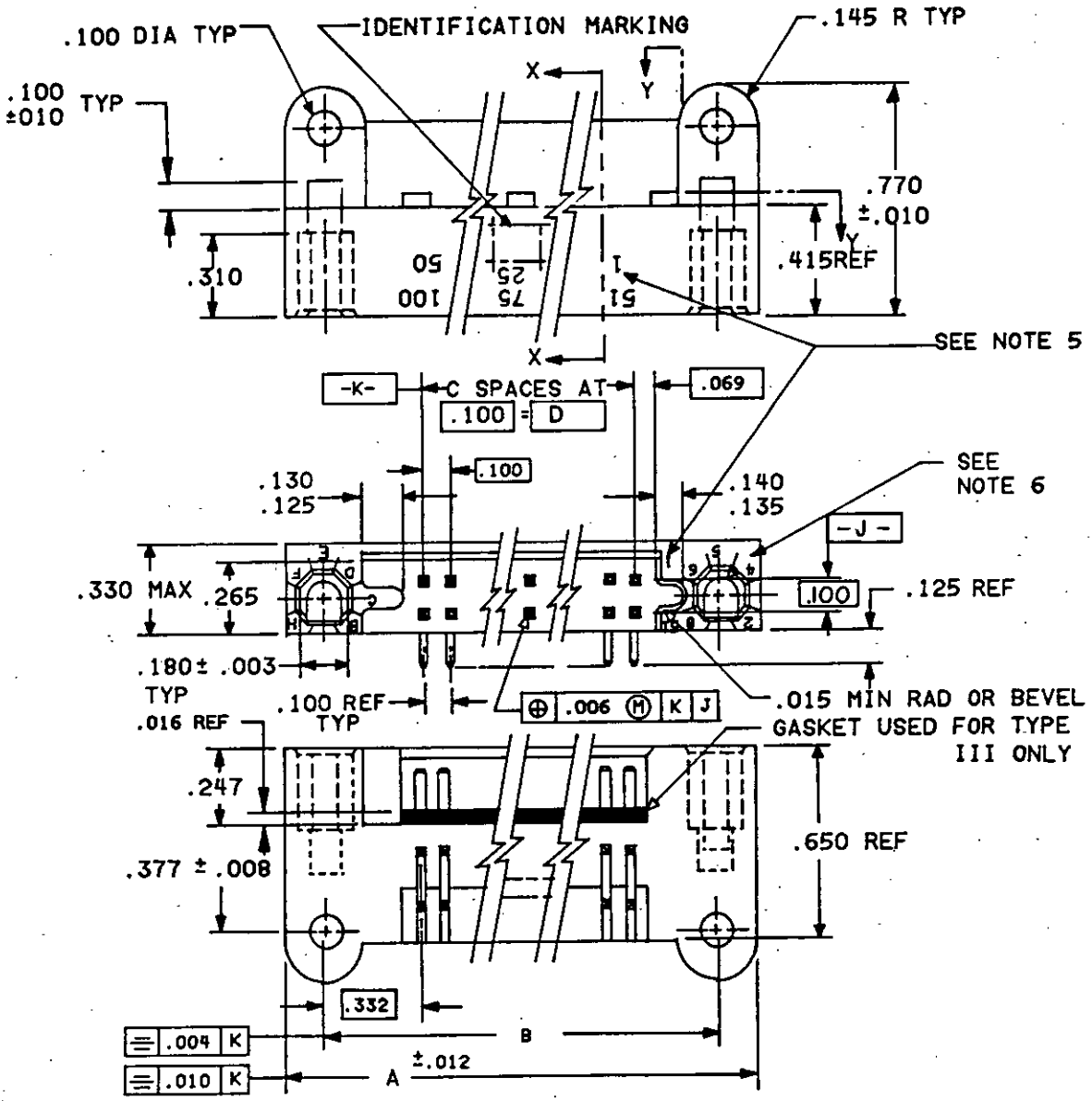



FIGURE 1. Connectors, plug (.100 sq. grid).

Ⓕ denotes changes

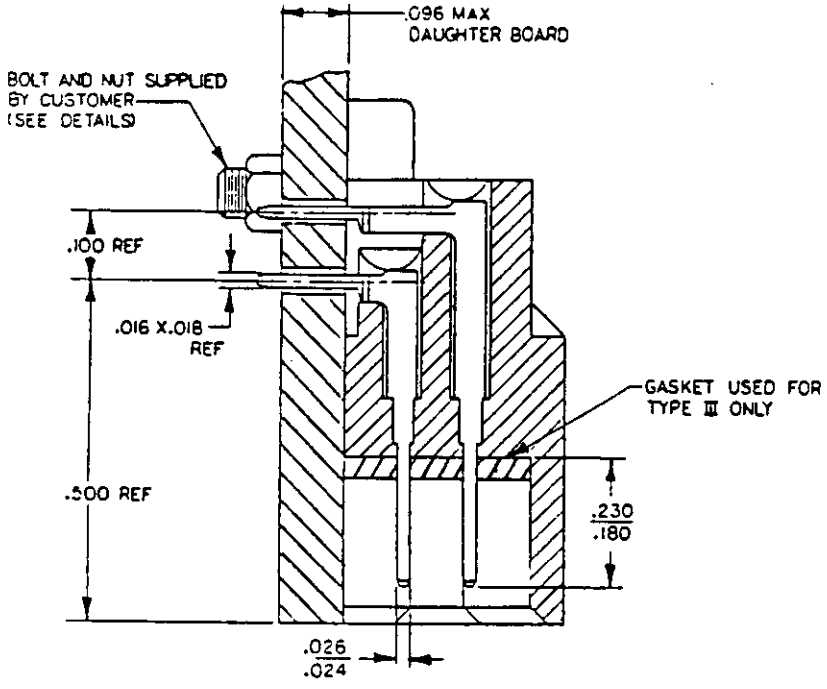


AMSC N/A
FSC 5935

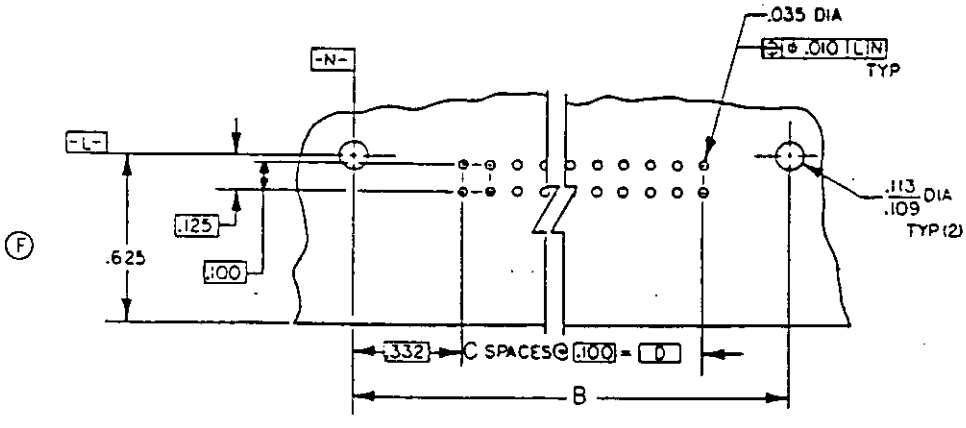
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Types II and III.




SECTION X-X

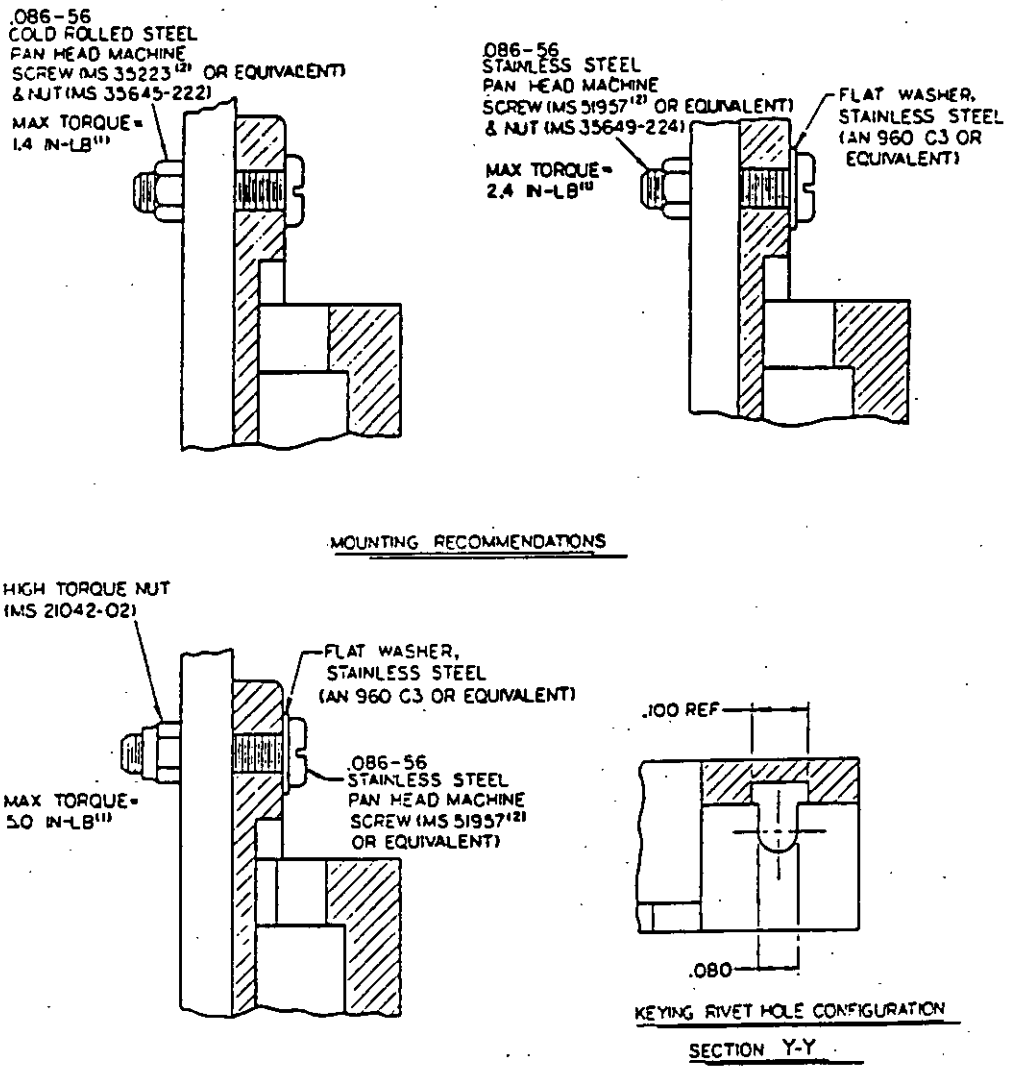



RECOMMENDED P.C. BOARD LAYOUT

FIGURE 1. Connectors, plug (.100 square grid) - Continued.



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NOTES:
1. Torque wrench must be used when fastening connector to printed circuit board.
2. Length to be determined by user.

FIGURE 1. Connectors, plug (.100 square grid) - Continued.



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Types II and III.

| Dash number 2/ | Type number | Number of contacts | Dimensions 1/ A | B | C | D | Contact identification numbers |
|---|---|---|---|---|---|---|---|
| 10<br>19 | II<br>III | 30 | 2.355<br>(59.82) | 2.065<br>(52.45) | 14 | 1.400<br>(35.56) | 1, 16 |
| 11<br>20 | II<br>III | 40 | 2.855<br>(72.52) | 2.565<br>(65.15) | 19 | 1.900<br>(48.26) | 1, 21 |
| 12<br>21 | II<br>III | 50 | 3.355<br>(85.22) | 3.065<br>(77.85) | 24 | 2.400<br>(60.96) | 1, 26 |
| 13<br>22 | II<br>III | 60 | 3.855<br>(97.92) | 3.565<br>(90.55) | 29 | 2.900<br>(73.66) | 1, 31 |
| 14<br>23 | II<br>III | 70 | 4.355<br>(110.62) | 4.065<br>(103.25) | 34 | 3.400<br>(86.36) | 1, 36 |
| 15<br>24 | II<br>III | 80 | 4.855<br>(123.32) | 4.565<br>(115.95) | 39 | 3.900<br>(99.06) | 1, 41 |
| 16<br>25 | II<br>III | 90 | 5.355<br>(136.02) | 5.065<br>(128.65) | 44 | 4.400<br>(111.76) | 1, 46 |
| 17<br>26 | II<br>III | 100 | 5.855<br>(148.72) | 5.565<br>(141.35) | 49 | 4.900<br>(124.46) | 1, 51 |
| 18<br>27 | II<br>III | 110 | 6.355<br>(161.42) | 6.065<br>(154.05) | 54 | 5.400<br>(137.16) | 1, 56 |

(F)

| Dash number | Type number | Number of contacts | Dimensions 1/ A | B | C | D | Contact identification numbers |
|---|---|---|---|---|---|---|---|
| 28<br>32 | II<br>III | 120 | 6.855<br>(174.12) | 6.565<br>(166.75) | 59 | 5.900<br>(149.86) | 1, 61 |
| 29<br>33 | II<br>III | 130 | 7.355<br>(186.82) | 7.065<br>(179.45) | 64 | 6.400<br>(162.56) | 1, 66 |
| 30<br>34 | II<br>III | 134 | 7.555<br>(191.90) | 7.265<br>(184.53) | 66 | 6.600<br>(167.64) | 1, 68 |
| 31<br>35 | II<br>III | 140 | 7.855<br>(199.52) | 7.565<br>(192.15) | 69 | 6.900<br>(175.26) | 1, 71 |

FIGURE 1. Connectors, plug (.100 square grid) - Continued.



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Types II and III.

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .003 | 0.08 | .058 | 1.47 | .140 | 3.56 |
| .005 | 0.13 | .069 | 1.75 | .180 | 4.57 |
| .006 | 0.15 | .072 | 1.83 | .230 | 5.84 |
| .008 | 0.20 | .096 | 2.44 | .247 | 6.27 |
| .010 | 0.25 | .100 | 2.54 | .265 | 6.73 |
| .012 | 0.30 | .101 | 2.57 | .310 | 7.87 |
| .015 | 0.38 | .109 | 2.77 | .323 | 8.20 |
| .024 | 0.61 | .113 | 2.87 | .332 | 8.43 |
| .025 | 0.64 | .125 | 3.18 | .377 | 9.58 |
| .026 | 0.66 | .130 | 3.30 | .415 | 10.54 |
| .030 | 0.76 | .135 | 3.43 | .500 | 12.70 |
| | | | | .770 | 19.56 |

1/ Metric equivalents are in parentheses.
2/ Supersession data:

| Type I superseded part number M55302/110- | Type II superseding part number M55302/110- |
|---|---|
| 01 | 10 |
| 02 | 11 |
| 03 | 12 |
| 04 | 13 |
| 05 | 14 |
| 06 | 15 |
| 07 | 16 |
| 08 | 17 |
| 09 | 18 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/27.
5. Numbers indicating every 5 cavities stamped on side. Numbers indicating end cavities and line indicating every 5 cavities stamped on mating face.
6. Key locations embossed on indicated surfaces.

FIGURE 1. Connectors, plug (.100 square grid) - Continued.



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REQUIREMENTS

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Material: In accordance with MIL-C-55302.

Gasket: Silicon rubber.

Ⓕ Plating: The contact plating for the engagement area, .150 inch minimum length, shall be gold over nickel per MIL-C-55302. The contact plating for the solder tail area, .160 inch minimum length, shall be tin lead over nickel per MIL-C-55302. The remainder of the contact shall be nickel plated per MIL-C-55302.

Contact identification: See figure 1.

Contact rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at 75°F.

Keying: See MIL-C-55302/31, using the M55302/31-10 rivet.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: 3 pounds minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

High altitude: 200 volts rms.

Part number: M55302/110- (and dash number from figure 1).

The Government has a royalty free license under the following patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

US Patent number 3,404,367

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - CR, AR, MI
Navy - AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3645-09)



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